Nazarbayev Intellectual Schools

Cambridge International Examinations
In collaboration with Nazarbayev Intellectual Schools, Kazakhstan

CANDIDATE NAME	

CENTRE NUMBER		CANDIDATE NUMBER	

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CHEMISTRY **Grade 12**

Paper 1 Multiple Choice **May 2014**

1 hour

Candidates answer on the Question Paper.

Additional Materials: Data Booklet
Electronic calculator

12CHEM/01

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number in the spaces at the top of the page.

DO **NOT** WRITE IN ANY BARCODES.
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct by putting a tick (✓) in the correct box.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working can be done in this booklet.
Electronic calculators may be used.

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This document consists of **16** printed pages.

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Section A

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For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct by putting a tick (✓) in the correct box.

1 *Use of the Data Booklet is relevant to this question*

Which statement about an oxygen atom in its lowest energy state is correct?

A The oxygen atom has eight electrons, all of which are paired.

B The oxygen atom has four filled orbitals of electrons.

C The oxygen atom has three filled orbitals of electrons.

D The oxygen atom has two full electron shells.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

2 Which equation represents the second ionisation energy of magnesium?

A $Mg^{+}(g) \rightarrow Mg^{2+}(g) + e^{-}$

B $Mg(g) \rightarrow Mg^{2+}(g) + 2e^{-}$

C $Mg^{+}(s) \rightarrow Mg^{2+}(s) + e^{-}$

D $Mg(s) \rightarrow Mg^{2+}(s) + 2e^{-}$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

3 Which statement correctly describes the attractive forces that occur within solid copper?

A The attractive forces are between anions and a sea of delocalised electrons.

B The attractive forces are between anions and cations.

C The attractive forces are between atomic nuclei and shared pairs of electrons.

D The attractive forces are between cations and a sea of delocalised electrons.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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4 Nickel chloride, $NiCl_2$, is an ionic compound. Phosphorus trichloride, PCl_3, is a covalent compound with a simple molecular structure. Nickel chloride and phosphorus trichloride have different properties, including different melting points and different electrical conductivity in the liquid state.

Which row of the table describes these differences correctly?

	melting point	electrical conductivity in the liquid state
A	$NiCl_2$ higher than PCl_3	$NiCl_2$ higher than PCl_3
B	$NiCl_2$ higher than PCl_3	PCl_3 higher than $NiCl_2$
C	PCl_3 higher than $NiCl_2$	$NiCl_2$ higher than PCl_3
D	PCl_3 higher than $NiCl_2$	PCl_3 higher than $NiCl_2$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

5 The mass spectrum of an organic compound, X, has a molecular ion peak with an m/z value of 80. It also has a peak with an m/z value of 81. The peak at 81 is smaller than the peak at 80, but its height is measurable.

What is the cause of the peak with an m/z value of 81?

A a new species that formed when two fragments of X combined together

B molecular ions of X that contain ^{13}C atoms

C molecular ions of X that contain ^{2}H atoms

D molecular ions of X that have gained one hydrogen atom

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

6 *Use of the Data Booklet is relevant to this question*

Carbon dioxide is a covalent compound with a simple molecular structure.

What is the total number of lone pairs, and of bond pairs, in the outer shells of the atoms in one carbon dioxide molecule?

	lone pairs	**bond pairs**
A	6	2
B	7	2
C	4	4
D	7	4

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

7 Which statement correctly describes the reaction that occurs, and the observation that would be made, when aqueous chlorine is added to aqueous potassium iodide?

	reaction	**observation**
A	chlorine is oxidised, iodide ions are reduced	a colour change is observed
B	chlorine is oxidised, iodide ions are reduced	no colour change is observed
C	chlorine is reduced, iodide ions are oxidised	a colour change is observed
D	chlorine is reduced, iodide ions are oxidised	no colour change is observed

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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8 *Use of the Data Booklet is relevant to this question*

Barium and magnesium are both group 2 metals which form hydroxide and sulfate compounds. These compounds have different solubilities in water.

Which statement is correct?

A $Ba(OH)_2$ is more soluble than $Mg(OH)_2$, $BaSO_4$ is more soluble than $MgSO_4$.

B $Ba(OH)_2$ is more soluble than $Mg(OH)_2$, $MgSO_4$ is more soluble than $BaSO_4$.

C $Mg(OH)_2$ is more soluble than $Ba(OH)_2$, $BaSO_4$ is more soluble than $MgSO_4$.

D $Mg(OH)_2$ is more soluble than $Ba(OH)_2$, $MgSO_4$ is more soluble than $BaSO_4$.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

9 *Use of the Data Booklet is relevant to this question*

A chemist makes up a solution of sulfuric acid. The solution contains 0.0981 g of sulfuric acid in every 10.0 cm^3 of solution.

What is the concentration of the solution?

A 0.000100 $mol\,dm^{-3}$

B 0.00981 $mol\,dm^{-3}$

C 0.100 $mol\,dm^{-3}$

D 9.81 $mol\,dm^{-3}$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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10 *Use of the Data Booklet is relevant to this question*

In a laboratory experiment, 4.215 g of magnesium carbonate was reacted with excess hydrochloric acid. After purification, 1.906 g of magnesium chloride was made.

What is the percentage yield of magnesium chloride?

A 40%

B 45%

C 52%

D 64%

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

11 The standard electrode potential of a half-cell can be found experimentally by using a standard hydrogen electrode. The conditions of the standard hydrogen electrode are important.

Two of the standard conditions are:

- electrical contact must be made by a platinum wire
- the H^+ concentration in the standard hydrogen electrode must be $1.0\,mol\,dm^{-3}$.

What are the other standard conditions?

A a temperature of 25 K and the use of a salt bridge made of calcium carbonate

B a temperature of 25 K and hydrogen gas at a pressure of 1.0 atm

C a temperature of 298 K and the use of a salt bridge made of calcium carbonate

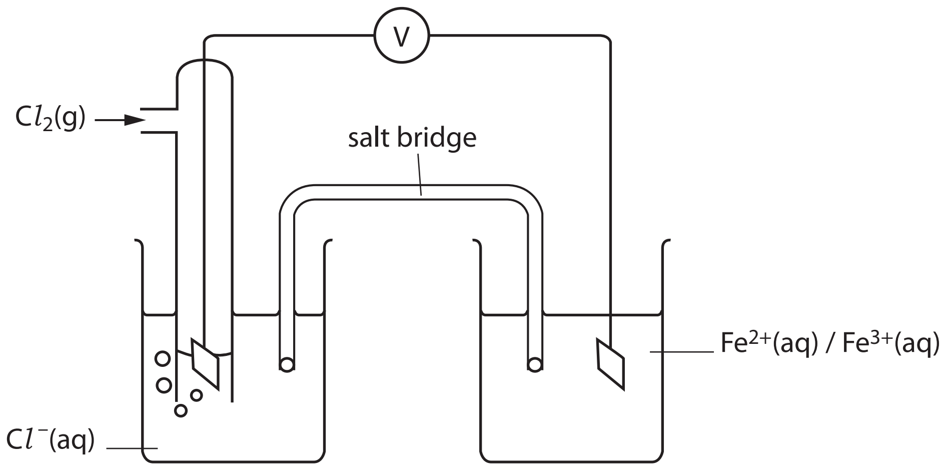
D a temperature of 298 K and hydrogen gas at a pressure of 1.0 atm

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

12 *Use of the Data Booklet is relevant to this question*

What is the cell reaction in the electrochemical cell shown below?



A $2Cl^- + Fe^{2+} \rightarrow Cl_2 + Fe$

B $3Cl_2 + 2Fe \rightarrow 6Cl^- + 2Fe^{3+}$

C $2Cl^- + 2Fe^{3+} \rightarrow Cl_2 + 2Fe^{2+}$

D $Cl_2 + 2Fe^{2+} \rightarrow 2Cl^- + 2Fe^{3+}$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

13 *Use of the Data Booklet is relevant to this question*

A student conducts an experiment in which he burns methanol, CH_3OH, to heat 80.0 cm^3 (80.0 g) of water. He burns exactly 1.00 g of methanol and the water is heated from 18.0 °C to 56.0 °C.

What experimental value for the enthalpy change of combustion of methanol can be calculated from this data?

A 5 kJ mol^{-1}

B 13 kJ mol^{-1}

C 407 kJ mol^{-1}

D 599 kJ mol^{-1}

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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14 The Haber process, $N_2 + 3H_2 \rightleftharpoons 2NH_3$, is catalysed by iron.

Which statement about the iron catalyst is correct?

A The iron is a heterogeneous catalyst that adsorbs the reactants onto its surface.

B The iron is a heterogeneous catalyst that raises the activation energy for the reaction.

C The iron is a homogeneous catalyst that adsorbs the reactants onto its surface.

D The iron is a homogeneous catalyst that raises the activation energy for the reaction.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

15 If nitrogen dioxide, NO_2, is placed in an evacuated glass bulb, dinitrogen tetroxide, N_2O_4, is formed. An equilibrium is established.

$$2NO_2\,(g) \rightleftharpoons N_2O_4\,(g) \qquad \Delta H^{\ominus} = -58\,\text{kJ}\,\text{mol}^{-1}$$

What change to the reaction conditions will increase the proportion of dinitrogen tetroxide at equilibrium?

A adding a platinum catalyst

B decreasing the pressure

C increasing the pressure

D increasing the temperature

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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16 Nitrogen monoxide, NO, reacts with oxygen to form nitrogen dioxide, NO_2.

$$2NO\ (g) + O_2\ (g) \rightarrow 2NO_2\ (g)$$

The rate equation for this reaction is:

$$\text{rate} = k[NO]^2[O_2]$$

In an experiment, measured concentrations of NO and O_2 gases were mixed and the initial rate of formation of NO_2 was found to be 0.00210 mol dm^{-3} s^{-1}. In a second experiment the concentration of NO gas was increased to three times its original value and the concentration of O_2 gas was increased to five times its original value.

What was the initial rate of formation of NO_2 in the second experiment?

A 0.0168 mol dm^{-3} s^{-1}

B 0.0315 mol dm^{-3} s^{-1}

C 0.0945 mol dm^{-3} s^{-1}

D 0.1575 mol dm^{-3} s^{-1}

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

17 A solution of sulfuric acid has a pH of 2.67.

What is the concentration of H^+ ions in the solution?

A 2.14 x 10^{-3} mol dm^{-3}

B 4.28 x 10^{-3} mol dm^{-3}

C 6.93 x 10^{-2} mol dm^{-3}

D 4.27 x 10^{-1} mol dm^{-3}

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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18 A student performed an accurate titration. She put $25.0\,cm^3$ of $0.100\,mol\,dm^{-3}$ aqueous sodium hydroxide solution into a conical flask, and added some indicator to it. Then she added hydrochloric acid to it until the indicator changed colour.

What apparatus should be used to measure the volumes used in the titration?

	to measure the sodium hydroxide	**to measure the hydrochloric acid**
A	bulb pipette	burette
B	conical flask	volumetric flask
C	measuring cylinder	burette
D	measuring cylinder	measuring cylinder

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

19 Titanium and aluminium have similar reactivity. Titanium ores are processed to produce titanium(IV) chloride, $TiCl_4$. Titanium is then extracted by reacting $TiCl_4$ with a reducing agent.

Which equation could represent the process used?

A $TiCl_4 + C \rightarrow Ti + CCl_4$

B $TiCl_4 + 2CO \rightarrow Ti + CCl_4 + CO_2$

C $TiCl_4 + 2Fe \rightarrow Ti + 2FeCl_2$

D $TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

20 Tin is an important metal, often used as a thin layer applied to steel, protecting the steel from corrosion. Ores of tin that can be used economically are running out.

What would cause the ores of tin to run out more quickly?

A developing alloys that can be used in place of tin

B developing additional uses for tin

C recycling the tin from used items

D using thinner layers of tin as corrosion protection

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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21 Bauxite is an important ore of aluminium. The bauxite is processed to obtain pure aluminium oxide which is then dissolved in molten cryolite. Aluminium is then obtained from this molten mixture by electrolysis.

Why is the aluminium oxide dissolved in molten cryolite?

A Aluminium oxide can act as an acid and as a base.

B Aluminium oxide is an ionic compound.

C Using molten cryolite prevents the oxidation of the carbon anodes.

D Using molten cryolite lowers the operating temperature.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

22 Structural isomerism **and** stereoisomerism should be considered when answering this question.

How many isomers of $C_4H_{10}O$ have an alcohol (OH) functional group?

A 2

B 3

C 4

D 5

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

23 The structural formula of butanal is $CH_3CH_2CH_2CHO$. Two samples of butanal are taken. Sodium metal is added to one sample. The other sample was added to Tollens' reagent, and the mixture was warmed.

What observations will be made in each case?

	butanal with sodium metal	**butanal with Tollens' reagent**
A	no visible change	no visible change
B	no visible change	silver metal is produced
C	rapid effervescence	no visible change
D	rapid effervescence	silver metal is produced

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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24 How many sigma, σ, and pi, π, bonds are there in one ethene molecule?

A no σ and 2 π

B 1 σ and 1 π

C 4 σ and 2 π

D 5 σ and 1 π

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

25 Which pair of substances will react together by a nucleophilic addition mechanism under suitable conditions?

A 1-bromobutane and aqueous sodium hydroxide

B butane and bromine

C butanone and hydrogen cyanide

D but-1-ene and bromine

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

26 Nitrobenzene can be made by treating benzene with a mixture of concentrated sulfuric and nitric acids at a temperature of 50 °C. In one step of the reaction an NO_2^+ ion interacts with a benzene molecule.

What happens in this step?

A a hydrogen atom breaks off the benzene ring and forms a bond to the NO_2^+ ion

B a pair of electrons from the benzene ring forms a bond to the NO_2^+ ion

C a pair of electrons from the NO_2^+ ion forms a bond to the benzene ring

D an uncharged intermediate is formed

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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27 Which statement correctly refers to the competitive inhibition of an enzyme?

A A competitive inhibitor binds to an enzyme away from the active site.

B A competitive inhibitor is unlikely to resemble the substrate.

C Competitive inhibition is reversible.

D V_{max} cannot be reached, even at high substrate concentration.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

28 Which statement about base pairing in DNA is correct?

A Each adenine base forms two hydrogen bonds to a guanine base.

B Each adenine base forms two hydrogen bonds to a thymine base.

C Each cytosine base forms two hydrogen bonds to a guanine base.

D Each cytosine base forms two hydrogen bonds to a thymine base.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

29 Which property of ATP enables it to perform one of its chief functions in biochemical systems?

A the hydrolysis of ATP is a highly endothermic reaction

B the hydrolysis of ATP is a highly exothermic reaction

C the molecules of ATP act as catalysts

D the molecules of ATP possess high kinetic energy

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

30 In which of the following biological situations or functions is either sodium metal or sodium ions involved?

A carrying oxygen around the body

B heavy metal poisoning by disrupting the tertiary structure of some enzymes

C the bonding that stabilises the α-helix in the secondary structure of proteins

D the transmission of nerve impulses

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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Section B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct.	1 and 2 only are correct.	2 and 3 only are correct.	1 only is correct.

No other combination of statements is used as a correct response.

31 *Use of the Data Booklet is relevant to this question*

Propane, propan-1-ol, and butanone are all analysed separately in a mass spectrometer.

Which of the mass spectra obtained have a peak due to a fragment with an m/z value of 29?

1 the mass spectrum obtained from propane

2 the mass spectrum obtained from propan-1-ol

3 the mass spectrum obtained from butanone

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

32 Which statements about bond polarity are correct?

1 Water molecules have polar bonds.

2 Giant covalent substances cannot have polar bonds.

3 Hydrogen molecules have polar bonds.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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33 Which statements about sulfur and its compounds are correct?

1 Sulfur can be obtained from deposits which contain the element sulfur itself.

2 Sulfur dioxide can be used as a food preservative.

3 Sulfur cannot exist in crystalline form.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

34 Ethanol can be made by a number of processes.

Which processes have an atom economy of at least 50%?

1 $C_2H_5Br + NaOH \rightarrow C_2H_5OH + NaBr$

2 $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

3 $C_2H_4 + H_2O \rightarrow C_2H_5OH$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

35 A student needs a buffer solution of pH 4.87. He has some propanoic acid ($K_a = 1.35 \times 10^{-5}$) and some sodium propanoate. He mixes them in water to make the buffer solution that he wants.

What **could** be the concentrations of propanoic acid and sodium propanoate in the buffer if its pH is 4.87?

1 propanoic acid 1.00 mol dm^{-3}, sodium propanoate 0.50 mol dm^{-3}

2 propanoic acid 0.50 mol dm^{-3}, sodium propanoate 0.50 mol dm^{-3}

3 propanoic acid 1.00 mol dm^{-3}, sodium propanoate 1.00 mol dm^{-3}

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

36 Which of these statements are correct?

1 K_w is the product of $[H^+]$ and $[OH^-]$.

2 K_b is the base dissociation constant.

3 pK_a is $-\log_{10}K_a$.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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37 Methane reacts with chlorine in the presence of ultra-violet light.

$$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$$

How can the first step in this reaction mechanism be described?

1 homolytic bond fission

2 the breaking of a Cl-Cl bond

3 a reaction between a molecule and a free radical

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

38 Which of these substances are condensation polymers?

1 nylon

2 polythene

3 polystyrene

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

39 Which of these formulae represent polyesters?

1 $[COC_2H_4COOC_2H_4O]_n$

2 $[COC_2H_4COOC_6H_4O]_n$

3 $[OCH(CH_3)CO]_n$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

40 Amylase is an enzyme found in saliva in the human mouth. Amylase catalyses the hydrolysis of starch.

Which process(es) could cause a sample of amylase to be denatured?

1 storing it at 80 °C

2 storing it at pH 13

3 storing it at 35 °C

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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